Question	Answers	Extra information	Mark	AO / Spec Ref.
04.4	tangent drawn to the 15 °C curve at 2 days	do **not** accept if there is an incorrect tangent at 2 days	1	AO2 4.7.2.3 RPA10
	$\frac{6.5 - 4.5}{4}$	allow a tolerance of ± ½ small square	1	
	0.5		1	
	$\frac{0.5}{0.3} = 1.67$	allow any number of decimal places allow answer to student's incorrectly calculated rate divided by 0.3 do **not** accept if a unit is given	1	

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.5	enzymes more active **or** more bacteria produced	allow enzymes work more quickly ignore enzymes work better	1	AO2
	lipids broken down more quickly	allow fats broken down more quickly	1	AO2
	<u>fatty acids</u> produced more quickly (which changes pH)	ignore glycerol do **not** accept incorrect products of lipid breakdown, eg amino acids or glycogen **alternative route** allow more (kinetic) energy (at higher temperature) (1) molecules move faster **or** more (successful) collisions **or** lipids broken down more quickly (1) <u>fatty acids</u> produced more quickly (which changes pH) (1)	1	AO1 4.2.2.1 4.7.2.3 RPA10

Total Question 4	**17**

Question 5

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.1		allow answers in terms of insulin for hormone **A** and / or glucagon for hormone **B** allow answers in terms of blood glucose (concentration) for substance **Q**		AO3 4.5.3.7 4.5.3.2 4.5.3.1
	if (concentration of substance) **Q** becomes high (hormone) **A** is released / used		1	
	if (concentration of substance) **Q** becomes low (hormone) **B** is released / used		1	
	hormone(s) / **A** / **B** brings (concentration of) substance **Q** back to ideal / normal		1	

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.2	(thyroxine) increases (basal) metabolic rate	allow (thyroxine) increases metabolism allow (thyroxine) increases respiration allow (thyroxine) increases reactions in cells	1	AO1 4.5.3.7 4.4.2.1 4.4.2.3
	<u>respiration</u> releases energy	allow <u>respiration</u> releases heat do **not** accept energy being produced / made / created allow <u>respiration</u> is exothermic	1	

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.3	(kidney) tubules less permeable to water		1	AO1
	(so) less water is reabsorbed	allow (so) less water taken back into the blood	1	AO1
	(so) more water in urine causing increased rate of production (of urine)	allow (so) more water in urine causing increased volume (of urine)	1	AO2
	(so) more water in urine causing lower concentration of dissolved substances (in urine)	(so) more water in urine causing more dilute urine	1	AO2 4.5.3.3

Total Question 5	**9**

Question 6

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.1	(0 to 6h): in the dark so (only) respiration occurs	ignore reference to animals allow no / limited light so (only) respiration occurs allow night(time) so (only) respiration occurs	1	AO2 4.7.1.2 4.7.2.2 4.4.1.2 4.4.2.1
	(respiration) produces carbon dioxide		1	
	(8 to 12h): in the light so photosynthesis (rate) is faster than respiration (rate)	allow in the light so more photosynthesis than respiration allow day(time) so more photosynthesis than respiration do **not** accept no respiration	1	
	idea of net / overall uptake of carbon dioxide	allow uptake of carbon dioxide by photosynthesis is greater / faster than output by respiration	1	

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.2	field of maize only takes in carbon dioxide for 6 months of the year (compared to all year)	answers must be comparative allow converse argument allow maize takes in carbon dioxide for a shorter amount of time	1	AO3 4.7.2.4 4.7.3.4 4.7.3.3
	(and) field of maize takes in less carbon dioxide per day than rainforest		1	

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.3	for comparison		1	AO2
	(because) different areas started with different numbers (of species)	do **not** accept (because) different areas started with different numbers of trees	1	AO3 4.7.2.4 4.7.3.1 4.7.3.4 4.7.3.6 4.7.3.3

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.4	any **one** from: • number of tree species increased rapidly at first • the number of tree species increases more slowly later • rate of increase of number of tree species decreases with time • beyond about 50 years there were more tree species than in the original rainforest • there was (great) variability of number of tree species from one study area to another at the same time	ignore answers referring to numbers of trees unqualified allow it takes 45 to 55 years for the forest to recover (the number of species) ignore biodiversity increases	1	AO3 4.7.2.4 4.7.3.1 4.7.3.6 4.7.3.3